Spoken Tutorial on Understanding APIs

Module	Module 3: An Introduction to Building with AI
Tutorial	1. Understanding APIs
Learning Objective	At the end of this tutorial learner will be able to 1. Define what an API is. 2. Explain the role of an API in AI applications. 3. Identify how an API facilitates interaction with AI models.
Approx. Duration	3-4 min
Outline	• Understanding API • API Analogy • Real-Life API Application • Direct AI Interaction • API-Driven AI Interaction • Comparing Interaction Methods
Meta Tags	API, Application Programming Interface, AI, Artificial Intelligence, Spoken Tutorial, Programming, Development, AI Models, Prompting, Interaction, Software Architecture
Pre-requisite Tutorial	Module 2: Introduction to Prompting

Script

Visual Cue	Narration
Title Slide	Welcome to this Spoken Tutorial on **Understanding APIs**.
Learning Objectives Slide	In this tutorial, you will learn to, • Define what an **API** is. • Explain the role of an **API** in **AI** applications. • Identify how an **API** facilitates interaction with **AI** models.
System Requirements Slide	Here I am using a browser on a computer or mobile.
Pre-requisite Slide	To follow this tutorial, you should have completed **Module 2: Introduction to Prompting**.

Illustration showing a customer, a waiter (API), and a kitchen (AI model) interacting in a restaurant setting	What is an **API**? An **API** is an **Application Programming Interface**. It acts as a messenger between different software. Think of an **API** like a waiter in a restaurant. You, the customer, are the application. You give your order, or **prompt**, to the waiter. The waiter, our **API**, takes your order to the kitchen. The kitchen is like an **AI model**. It prepares your food, or the **AI** output. The waiter then brings your food back to you. This is how an **API** helps applications talk to **AI**.
Screen close-up: user typing 'tomato...' in a grocery app search bar on a smartphone	Let's see an **API** in action with a real-life example. Open any grocery app on your phone. Tap the search bar at the top. Type 'tomato' and press search. Behind the scenes, your app sends a request to the **API**. The **API** communicates with the grocery store's server. It finds all the tomato products available. The server sends back this list of results to the **API**. The **API** then delivers these results to your app. You now see all tomato products on your screen. This whole process uses an **API**.
Illustration showing a person looking confused amidst complex code and diagrams, representing direct AI interaction	Imagine you want to use an **AI model**. One way is to interact with it directly. This means going deep into its internal code. You would need to understand all its complex parts. Think of it like trying to cook your own meal in a restaurant kitchen. You need to know where everything is kept. You also need to understand all the cooking steps. This direct method is very complicated and error-prone. It also needs deep technical knowledge. This is not practical for most applications.
Illustration showing a person easily ordering from a menu with a waiter, representing simplified API-driven AI interaction	Now, let's look at the **API** approach. This is the improved and easier way to interact with **AI**. The **API** acts as a clear, simple interface. It handles all the complex communication for you. Think of it like ordering from a menu with a waiter. You simply tell the waiter what you want. The waiter, our **API**, takes care of everything else. It ensures your request reaches the **AI model** correctly. The **API** also makes the interaction secure and efficient. This standardization saves a lot of time and effort.

Side-by-side comparison: complex direct interaction (person making bricks) vs. simplified API interaction (person using ready-made bricks for building)	Let us compare direct interaction versus **API** interaction. Direct interaction is complex and prone to errors. It requires deep knowledge of the **AI model**'s internals. On the other hand, **API** interaction is simple. The **API** provides a standardized way to communicate. Imagine building a house. Direct interaction is like making every single brick yourself. Using an **API** is like buying ready-made bricks. The **API** method offers better security and scalability. It makes developing **AI** applications much faster.
Summary Slide	Let us summarize what we learned. An **API** is an **Application Programming Interface**. It acts as an intermediary for software communication. We used a restaurant analogy to understand its function. We saw how **API**s work in a grocery app example. **API**s simplify interacting with complex **AI** models. They offer a more secure and standardized approach.
Assignment Slide	Now as an assignment, Identify three apps on your phone that likely use **API**s. For each app, describe one interaction that involves an **API** request. Compare the results.
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